Al- Balqa' Applied University
Faculty of Engineering Technology
Basic Science Department
Department of Physics and Basic Sciences

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم الفيزياء والعلوم الأساسية

Mid Exam

Course Title	General physics lab.1		Course Code	30201111	
Semester	Second		Academic Year	2025-2026	
Exam Date	4-5-2026	Day		Time	14:30-15:20

Student Name		Student ID No.		Section	

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Essay	CLIO, 2,3,4,5,6,7	(36.67%)	11		
2	Essay	CLIO, 2,3,4,5,6,7	(30.00%)	9		
3	Multiple Choice	CILO 1,6,8	(8.33%)	2.5		
4	Multiple Choice	CILO 1,6,8	(8.33%)	2.5		
5	Multiple Choice	CILO 1,6,8	(6.67%)	2		
6	Multiple Choice	CILO 1,6,8	(3.33%)	1		
7	Multiple Choice	CILO 1,6,8	(6.67%)	2		
Total			(100%)	30		

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

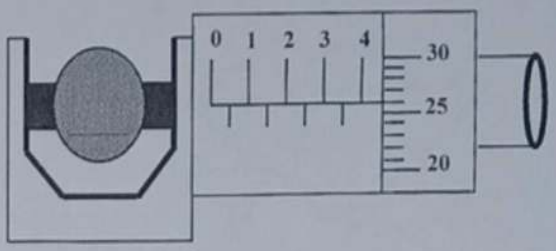
T.C.N.E
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*Note: When you need consider the acceleration of gravity (g) =10 m/s^2, $\pi = 3.14$

Q1: Measurements and Errors

The diameter of a small sphere was measured by the micrometer shown in the figure beside.



1) What is the reading of the micrometer (in **mm**)..... 4.26 ..... **[2 points]**

2) Calculate the volume of the sphere (in **cm^3**) **[2 points]**

$V = \frac{4}{3}\pi\left(\frac{d}{2}\right)^3$ or $= \frac{4}{3}\pi r^3 = \frac{4}{3} * 3.14\ (0.213)^3$

$V = 0.0405\ cm^3$

3) Calculate the uncertainty in the calculated volume, (ΔV) (in cm^3) **[3 points]**

$\frac{\Delta V}{V} = 3\frac{\Delta d}{d}$ or $= 3\frac{\Delta r}{r}$

$= 3 * \frac{0.005}{4.26} = 3 * \frac{0.0025}{2.13} = 3.521 * 10^{-3}$

$\Delta V = 3.521 \times 10^{-3} * V = 3.521 * 10^{-3} * 0.0405 = 1.426 \times 10^{-4}\ cm^3$

4) IF the mass of the sphere **M = 0.6 g** was measured by a digital balance, **then** calculate the density of the sphere (in **g/cm^3**). **[2 points]**

$\rho = \frac{M}{V} = \frac{0.6}{0.0405} = 14.81\ g/cm^3$

5) If the density is 14.56 g/cm^3, calculate the percentage error in the calculated density of the sphere.

$P.E = \frac{|\rho_{exp} - \rho_{th}|}{\rho_{th}} \times 100\%$ **[2 points]**

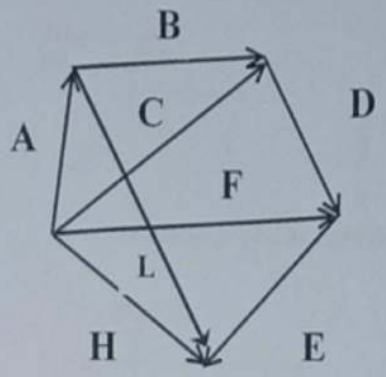
$= \frac{|14.81 - 14.56|}{14.56} * 100\% = 1.72\%$

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Q2: Force Table Experiment (Vectors):

I) In the Figure shown beside;
Which vector is the resultant of **C+D+E** ?

Answer: Vector---- H ---- **[1 point]**



II) **Three forces are in equilibrium. Calculate the missing quantities depending on the data given in table below. (show your calculations in details): [8 points]**

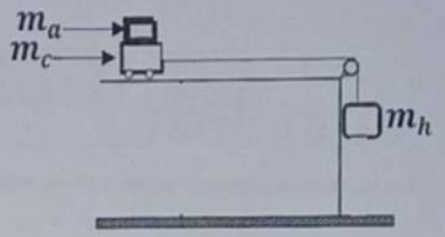
[Note: The angles (Θ_1, Θ_2 and Θ_3) are with (positive x-axis counterclockwise)

Mass (g)	Vector (N)	Θ^0 $\tan^{-1}\frac{Fy}{Fx}$	X-component $F\cos\theta$	Y-component $F\sin\theta$
? $2.5 = m \times 10$ $m_1 = 0.25\ kg$ $= 250\ g$	F_1= ? $\frac{F_{1x}}{\cos 0} = 2.5$	0^0	2.5	? $2.5 \sin 0 = 0$
120	F_2 $0.120 \times 10 = 1.2$ نرفضها	165^0	? $1.2 \cos 165$ -1.159	? $1.2 \sin 165$ 0.311
	F_3	? $\tan^{-1} -0.311 = 13.06$ $\theta_R = 13.06 + 180 = 193.06$	? $\Sigma F_x = 0$ $2.5 - 1.159 - F_{3x} = 0$ $F_{3x} = -1.341$ الاشارة	? $\Sigma F_y = 0$ $0 + 0.311 + F_{3y} = 0$ $F_{3y} = -0.311$ الربع

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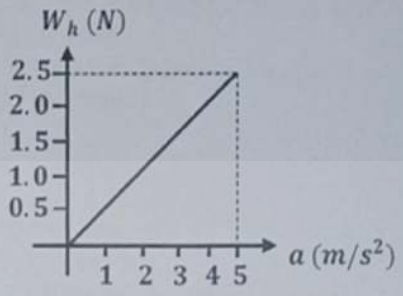
Force and Motion (Newton's Second Law):



The relation between the weight of a hanging mass, W_h , and the acceleration, a , of the system is shown in Figure beside.

Q3: The **total mass** of the system is (**in Kg**) **[2.5 points]**

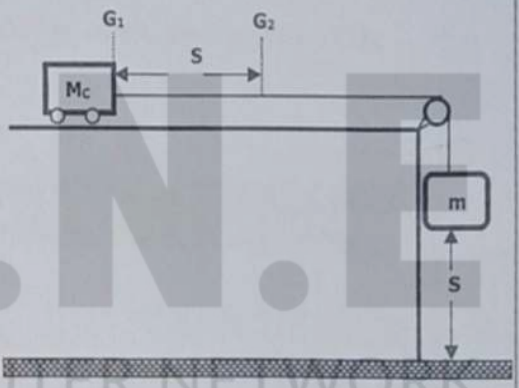
A) 1.2 B) 0.4 C) 0.5 D) 2.0 E) 0.7



Q4: Assume the cart mass is **0.15 kg**, At the instant $a = 3m/s^2$, the added mass m_a , (in Kg) is: **[2.5 points]**

A) 0.2 B) 0.15 C) 0.35 D) 0.10 E) 0.25

Friction:



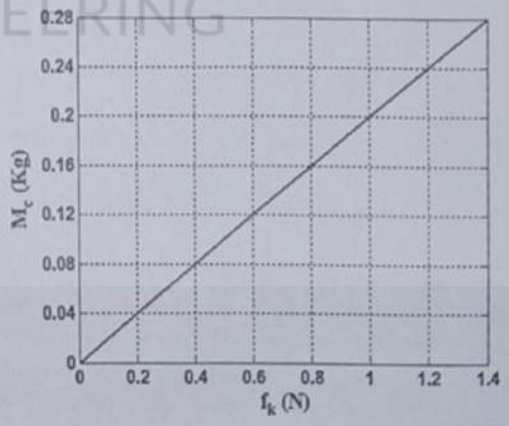
Q5: The relation between the normal force (n) acts on mass m and the kinetic friction force (f_k) between m and the air track surface is shown beside. calculate the **kinetic friction coefficient** (μ_k)? **[2 points]**

A) 0.7 B) 0.3 C) 0.2

D) 0.4 E) 0.5

Q6: Complete the following statement:
The unit of the slope is Kg/N which is equivalent to :

..... s^2/m ..... **[1 points]**



Q7: If $m_h = 0.15$ Kg and $M_c = 0.12$ Kg, calculate the **acceleration** of the system, in m/s^2?
[2 points]

A) 3.78 B) 4.67 C) 2.57

D) 3.33 E) 7.43

With Best Wishes

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